

AARON DUTTA

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EDUCATION

University of Waterloo

BMath, Honours Mathematics with Co-Op

Expected June 2031

Waterloo, ON

SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML5/CSS3, SQL

Libraries/Frameworks: FastAPI, Flask, Next.js, Prisma, React, Streamlit, Tailwind CSS

Developer Tools: Anthropic API, Docker, Figma, Git, MediaPipe, NetworkX, NumPy, OpenCV, Pandas, PostgreSQL, Pydantic, RESTful APIs, SQLite, Supabase, Vercel

EXPERIENCE

Filix School of Education

2024 - Present

Mathematics Mentor

- Mentoring **Grade 7-12** students through weekly **math circles**, encouraging collaborative problem solving
- Advancing students' **critical thinking** skills by combining theoretical concepts with practical exercises

Olympian Swimming

2023 - Present

Swim Coach

- Delivering structured swim lessons focused on **skill development**, swim posture, and stroke technique
- Maintaining attendance and evaluating progress through comprehensive **report cards** to track growth

PROJECTS

TechniqueTitan

2026

Python, MediaPipe, OpenCV, FastAPI, React, TypeScript, NumPy



- Engineered a **MediaPipe + OpenCV + NumPy** pipeline that extracts **21 landmarks** per hand, normalizes for camera distance, and scores **5 posture criteria**; hardened laterality with bilateral **hand disambiguation** and a **45-test pytest** suite across geometry, scoring, and API modules
- Shipped a **FastAPI + React/TypeScript** product (photo, video timeline, live camera) with on-device browser MediaPipe and YAML-tunable thresholds, cutting review from a full lesson cycle to seconds
- Built a **batch CLI** that walks image folders and exports **CSV/JSON** metrics plus outlier flags, automating dataset-scale analysis that was previously manual

SurplusLink

2nd Best Overall, KYHacks 2026

Next.js, TypeScript, React, Tailwind, Prisma, Supabase, Vercel



- Won **2nd Best Overall** at **KYHacks 2026** for a phone-first surplus-food marketplace connecting restaurant donors to nearby recipients before pickup windows close
- Built a **Next.js** donor camera flow with a local **Food-101 ONNX** classifier from Hugging Face Transformers suggesting title, categories, and allergen heuristics
- Shipped recipient map/list claims with **2-opt** multi-stop routing, transactional portion locks, and a **Vercel + Supabase** production deploy

FraudGen

2026

Python, Anthropic API, FastAPI, Streamlit, Pandas, NetworkX, Pydantic



- Designed a **4-agent adversarial LLM** pipeline that explores fraud variant space via a **coverage matrix** and generates **persona-seeded mule-network** scenarios
- Shipped **async parallel orchestration**, deterministic **8-rule** validation/repair, and **critic scoring** with retry loops; a single produced variant labeled **11 transactions** across a 3-victim → 4-mule converging funnel with 12 evasion techniques
- Built a **Streamlit** monitoring console with **network graph** visualization and **ML-ready exports** for **XGBoost/GNN** training, and exposed the pipeline through **FastAPI REST** endpoints

CERTIFICATIONS

Essentials with Azure Fundamentals, Microsoft

2026

Google AI Essentials Specialization, Google

2026

Building with the Claude API, Anthropic

2026

CS50x, Harvard University

2024